AQA 3.1.4 and 3.1.10 Equilibrium constants K_c and K_p

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- Partial pressure = mole fraction × total pressure
- Find equilibrium amounts with a start / change / equilibrium table using the mole ratio in the equation
- K_c: divide each amount by the volume in dm^3 before substituting (unless the volumes cancel)
- Only temperature changes the value of K_p or K_c; pressure, concentration and catalysts do not
- Exothermic forward reaction: raising temperature lowers K; endothermic: raising temperature increases K

Section A: Short-answer questions

01 For $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, $K_c = 4.0$ at a certain temperature. Calculate K_c for $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ at the same temperature.

[1 mark]

02 An equilibrium mixture is prepared in a container of fixed volume.

$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g) \quad \Delta H = -108 \text{ kJ mol}^{-1}$

The temperature of this mixture is decreased and the mixture is allowed to reach a new equilibrium. Which is greater for the new equilibrium than for the original equilibrium?

[1 mark]

A The mole fraction of carbon monoxide
B The partial pressure of chlorine
C The total pressure of the mixture
D The value of the equilibrium constant, K_p

03 Which statement about catalysts used in reactions at equilibrium, at a constant temperature, is correct?

[1 mark]

A They are always used in the solid state.
B They increase the rate of the forward reaction but decrease the rate of the reverse reaction.
C They have no effect on the value of the equilibrium constant.
D They make the forward reaction more exothermic.

04 The mole fraction of nitrogen in an equilibrium mixture is 0.30 and the total pressure is 150 kPa. Calculate the partial pressure of nitrogen.

[1 mark]

Section B: Longer questions

05 $A(aq) + 2B(aq) \rightleftharpoons C(aq) \quad \Delta H = -32 \text{ kJ mol}^{-1}$

05.1 A solution containing 0.60 mol of A is added to a solution containing 0.60 mol of B. The amount of C formed at equilibrium is 0.28 mol. Deduce the amounts, in moles, of A and B in this mixture at equilibrium.

[2 marks]

05.2 Give an expression for the equilibrium constant (K_c) for this reaction.

[1 mark]

05.3 The temperature of the equilibrium mixture is decreased. Predict the effect, if any, on the value of K_c. Give a reason for your prediction.

[3 marks]

05.4 In another mixture at equilibrium [A] = 0.48 mol dm^{-3} and [C] = 0.62 mol dm^{-3}. For this reaction, K_c = 7.8 mol^{-2} dm^6. Calculate [B] at equilibrium. Give your answer to the appropriate number of significant figures.

[3 marks]

06 Nitrogen monoxide reacts with chlorine to form nitrosyl chloride (NOCl).

$2NO(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$

06.1 1.50 mol of NO are mixed with 1.00 mol of Cl_2 and the mixture is left to reach equilibrium at a given temperature. The equilibrium mixture contains 0.350 mol of NOCl. Calculate the amount, in moles, of NO and of Cl_2 in the equilibrium mixture.

[2 marks]

06.2 Give the expression for the equilibrium constant, K_c, for this reaction.

[1 mark]

06.3 A different equilibrium mixture is prepared in a flask of volume 800 cm^3 at a different temperature. At equilibrium this mixture contains 0.850 mol of NO and 0.458 mol of Cl_2. For the reaction at this temperature $K_c = 1.32 \times 10^{-2}$ mol^{-1} dm^3. Determine the amount, in moles, of NOCl in this equilibrium mixture.

[4 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	0.25	*Source: Practice question (written for this set)*	1
02	**D**	*Source: 7405/3 Jun 2017 Q20*	1
03	**C**	*Source: 7405/3 Jun 2023 Q25*	1
04	45 kPa	*Source: Practice question (written for this set)*	1
05.1	A = 0.32 mol B = 0.04 mol	*Source: 7404/1 Jun 2023 Q08*	2
05.2	$K_c = [C] \div [A][B]^2$		1
05.3	M1 greater / increases M2 the equilibrium shifts in the exothermic direction M3 to increase the temperature / oppose the temperature decrease		3
05.4	M1 $[B]^2 = [C] \div ([A]\ K_c) = 0.62 \div (0.48 \times 7.8)$ M2 [B] = 0.4069 M3 [B] = 0.41 (2 sf)		3
06.1	NO = 1.15 mol Cl_2 = 0.825 mol	*Source: 7404/1 Jun 2019 Q05*	2
06.2	$K_c = [NOCl]^2 \div [NO]^2[Cl_2]$		1
06.3	M1 divides mole quantities by 0.800 M2 $[NOCl]^2 = 8.53 \times 10^{-3}$ mol^2 dm^{-6} M3 [NOCl] = 0.0924 mol dm^{-3} M4 n(NOCl) = 0.0924 × 0.800 = 0.0739 mol	If no division in M1 then max 3 (M4 = 0.0529 mol)	4